GGG-L-211D
August 27, 1987
SUPERSEDING
GGG-L-211C
October 30, 1967

FEDERAL SPECIFICATION

LEVELS AND PLUMBS, AND LEVELS

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers levels, and levels and plumbs.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, and styles. Levels and levels and plumbs covered by this specification shall be of the following types, classes, and styles, as specified (see 6.2):

Levels and plumbs:

- Type I. Carpenter's.
  - Class 1. Metal, siding reading.
    - Style A. Adjustable vial.
    - Style B. Fixed vial.
  - Class 2. Wood, top reading.
    - Style A. Adjustable vial.
    - Style B. Fixed vial.
  - Class 3. Wood, side reading.
    - Style A. Adjustable vial.
    - Style B. Fixed vial.
- Type II. Bench, precision (level and plumb).
  - Class 1. Single face, fixed vial.
    - Style A. Machinery.
  - Class 2. Single face, adjustable vial.
    - Style A. Machinist's.
    - Style B. Millwright's.
- Type III. Mason's.
  - Class 1. Metal, side reading.
    - Style A. Adjustable vial.
    - Style B. Fixed vial.
  - Class 2. Wood, side reading.
    - Style A. Adjustable vial.
    - Style B. Fixed vial.
- Type IV. Torpedo (pocket).

Levels:

- Type VI. Bench (level).
  - Class 1. Single face.
    - Style A. Fixed vial.
  - Class 2. Straightedge.
  - Class 3. Cross test.
  - Class 5. Striding.

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Type VII. Midget (pocket).

Type VIII. Line.

Type X. Circular level (mounted).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specification, of the issues in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions as outlined under General Information in the Index of Federal Specifications and Standards. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the General Services Administration, Specifications Unit (WFSIS), 7th and D Street SW, Washington, DC 20407.)

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Commercial Item Descriptions and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of levels and plumbs, and levels which are otherwise in accordance with this specification.

3.2 Design. The levels and plumbs, and levels shall be so designed as to satisfactorily meet the requirements of this specification for the type, class, and style specified (see 6.2).

3.3 Definition. A level is an instrument or device for determining the truth of a horizontal plane or any deviation therefrom. A plumb, as a component part of a level and plumb, is an instrument or tool for establishing the truth of a plane perpendicular to the aforementioned horizontal plane or any deviation from same.

3.3.1 Accuracy. The level accuracy is the measure of the allowable deviation or tolerance on the level reading itself. It is determined by the setting of the vial within the stock as checked against a true horizontal or vertical plane within tolerances set forth herein.

3.3.2 Vial sensitivity. Measurement of vial sensitivity is the amount of angular movement in a vertical plane required to move the vial bubble 1/10 inch. Angular movement may be expressed in degrees, minutes, or seconds. The sensitivity

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of the bubble motion in the vial is governed by the vial radius; the larger the radius, the smaller the angle required to move the bubble 1/10 inch.

3.4 Material. Material shall be as specified hereinafter.

3.5 Tolerances.

3.5.1 Vial setting. Vial setting (see letter "a" of figure 1), shall not exceed the tolerances specified herein for the levels and plumbs, and levels specified for each utile surface.

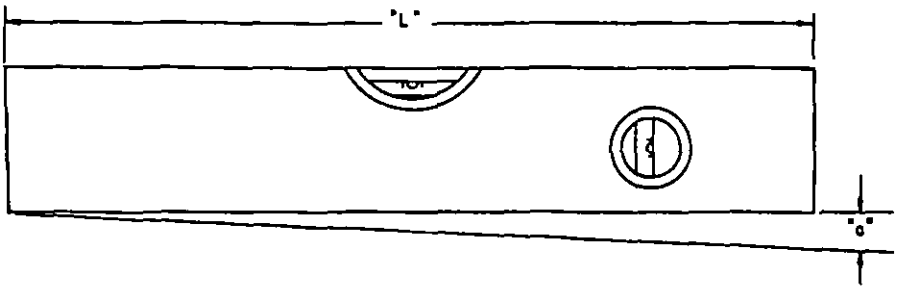


FIGURE 1. Vial bubble setting tolerance.

3.5.2 Concavity and convexity. Level concavity (see letter "b" of figure 2), shall not exceed the tolerances specified herein for the levels and plumbs, and levels specified. Convexity of the utile surfaces shall not exceed 0.002 inch.

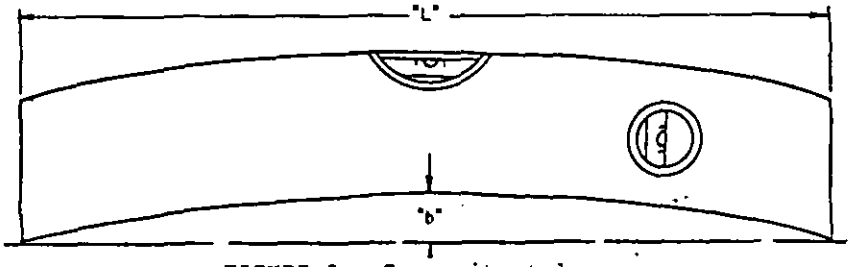


FIGURE 2. Concavity tolerance.

3.5.3 Parallelism. Level parallelism (see letter "c" of figure 3), shall not exceed the tolerances specified herein for the levels and plumbs, and levels specified.

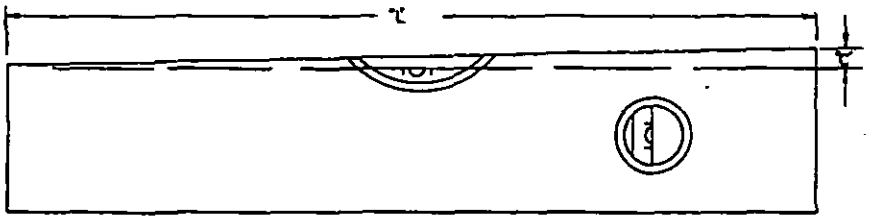


FIGURE 3. Parallelism tolerance (not applicable to single faced levels).

NOTE: Tolerances shown on figures 1, 2, and 3 greatly exaggerated for clarity.

3.6 Stocks. Level and plumb, and level stocks shall be made of the material specified hereinafter and shall be designed to prevent warping, to properly support vials or vial holders, and shall be within the specified tolerances. End plates and vial protectors shall be solid aluminum, aluminum alloy, brass or steel and shall be fitted so as to cause no interference with utile surfaces. Hand grooves shall be of an optimum depth. A handhold in accordance with the manufacturer's standard shall be provided in wood stocks.

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3.7 Vials. The vials shall be mounted so that the bubble is clearly visible and shall be filled with a nonfreezing liquid solution equivalent in viscosity to 25 percent ether or 50 percent alcohol. The liquid shall be either clear or amber colored (Cats Eye), at the option of the manufacturer, and shall not discolor or cloud or change in fluidity with age or at a temperature as low as -20 degrees and as high as +120 degrees F for types II and VI levels and +150 degrees F for types I, III, IV, VII, and VIII levels. The liquid shall form a well-defined bubble of a size to lie just within the inside edge of the defining centering lines of the vial.

3.7.1 Fixed vials. The fixed vials shall be firmly and permanently set in the stock of the level.

3.7.2 Adjustable vial holders. Adjustable vial holders shall properly support and provide adjustment and protection for the vials. Vial holders shall be of a size and design to be interchangeable and replaceable in a level of the same type, class, style, and size of any one manufacturer.

3.7.3 Vial lenses. Vial lenses shall be of glass or plastic that will not excessively scratch or become marred or discolored under usage. Lenses shall be held firmly in place and shall not become loose in normal use.

3.8 Finish.

3.8.1 Wood parts. Wood parts shall be finished smooth and shall be coated with varnish, shellac, lacquer, or other material to prevent moisture absorption.

3.8.2 Metal parts.

3.8.2.1 Utile surfaces. Surfaces employed in the utility of the level shall have a surface roughness not to exceed 125 microinches (arithmetical average) A.A., except for flat surfaces on type II, which shall be 32 microinches A.A. The spotting cavities, if present, shall be from 0.0002 to 0.0003 inch in depth with a concentration of 10 to 12 spotting cavities per square inch (p.s.i) and a 20 to 40 percent bearing. Levels shall have no burrs, fins, sharp projections, excessive surface roughness, protruding end or cover plates, or any imperfections which might impair their serviceability.

3.8.2.2 Non-utile surfaces, machined. Machined surfaces not employed in the utility of the level shall have a surface roughness not exceeding 250 microinches A.A.

3.8.2.3 Non-utile surfaces, unmachined. Unmachined ferrous metal surfaces shall be smooth. These surfaces shall be japanned, lacquered, or coated with other suitable protective coating. Nonferrous metal surfaces may or may not be coated.

3.8.3 Component parts. All aluminum, aluminum alloy, and brass parts shall be smoothly finished. All steel parts shall be electroplated or painted in accordance with manufacturer's commercial standards.

3.9 Type I, carpenter's.

3.9.1 Vials. In addition to conforming to the above, the level and plumb vials for all type I carpenter's levels and plumbs shall be commercial bent-glass tubes and shall be permanently marked with two black bubble-defining centering lines. The vials shall be bent to a radius of 7.645 inches, minimum, equivalent to a vial sensitivity of 45 minutes maximum, and shall be capable of passing the tests in section 4 for type I levels. The radius of curvature shall be uniform along the length of the vial body.

3.9.2 Class 1, metal, side reading.

3.9.2.1 Accuracy. Class 1 levels and plumbs shall conform to the tolerances of table I for the accuracy for vial setting "a", concavity "b" and parallelism "c" as illustrated in figures 1, 2, and 3 respectively. The levels shall conform to the convexity requirements of 3.5.2. The levels and plumbs shall be capable of passing the tests specified in section 4, as applicable.

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TABLE I. Tolerances for levels and plumbs, and levels with 45 minute vials.

Length of level "L"	Maximum tolerance	
	a	b or c
6 or less	0.003	0.005
8	.004	.005
12	.007	.005
18	.011	.010
24	.014	.010
26	.015	.010
28	.017	.010
30	.018	.010
48	.026	.015

3.9.2.2 Style A, adjustable vial. The adjustable vial, metal, carpenter's levels and plumbs shall consist essentially of a stock, a twin level vial, two single or twin plumb vials, and a corresponding number of vial holders. They shall conform to table II for the length specified (see 6.2), and shall be similar to figure 4.

TABLE II. Type I, class 1, style A, adjustable vial, metal, side reading, carpenter's levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
18	2-1/4	2-5/8	1	1-1/8	1-7/8
24	2-1/4	2-5/8	1	1-1/8	2-3/8
28	2-1/4	2-5/8	1	1-1/8	2-3/4

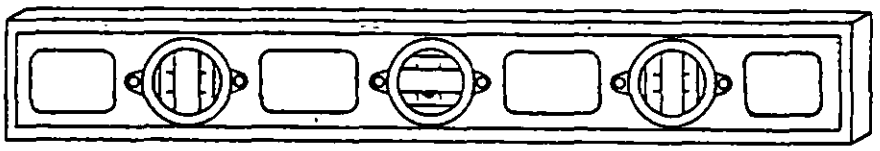


FIGURE 4. Type I, class 1, style A, adjustable vial, metal, side reading, carpenter's level and plumb.

3.9.2.2.1 Stock. Levels and plumbs of this style shall have a stock of aluminum alloy of a truss-type or modified channel design. The stock shall be designed to provide strength and rigidity, and shall pass the loading test requirements of 4.5.3.6.

3.9.2.2.2 Vials. Level vials and plumb vials shall be adjustable.

3.9.2.2.3 Vial holders. Vial holders shall be of aluminum, aluminum alloy, zinc alloy, steel or plastic so mounted as to assure side reading. Steel parts shall be coated to prevent corrosion.

3.9.2.3 Style B, fixed vial. Style B, fixed vial, metal, side reading, carpenter's levels and plumbs, shall conform to the requirements for type 1, class 1, style A, except that the level vials and plumb vials shall be fixed. The level shall have a twin level vial and two twin plumb vials. The vials shall be secured in the cavity of the frame in such a way as to be fixed and nonadjustable parts of the level assembly. They shall conform to table III for the length specified (see 6.2) and shall be similar to figure 5.

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TABLE III. Type I, class 1, style B, fixed vial, metal, side reading, carpenter's levels and plumbs.

Length nominal	Depth		Width Minimum	Weight Maximum
	Minimum	Maximum		
Inches	Inches	Inches	Inch	Pounds
12	2-1/8	2-5/8	1	1-1/2
18	2-1/8	2-5/8	1	1-3/4
24	2-1/8	2-5/8	1	2-1/4
28	2-1/8	2-5/8	1	2-1/2
30	2-1/8	2-5/8	1	2-3/4

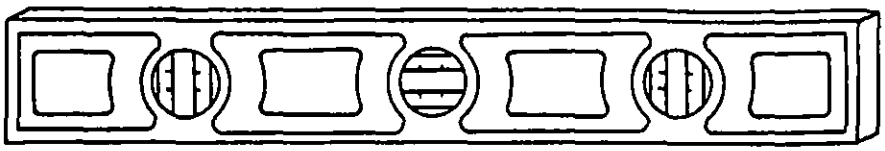


FIGURE 5. Type I, class 1, style B, fixed vial, metal, side reading, carpenter's level and plumb.

3.9.3 Class 2, wood, top reading.

3.9.3.1 Accuracy. Class 2 levels and plumbs shall conform to the tolerances of table I for vial setting dimension "a" as illustrated by figure 1, the convexity requirements of 3.5.2 and the concavity, dimension "b" as illustrated by figure 2 shall not exceed 0.015 inch total indicator variation when tested in accordance with 4.5.2.1. The levels and plumbs shall be capable of passing the tests specified in section 4, as applicable.

3.9.3.2 Style A, adjustable vial. Style A, adjustable vial, wood, top reading, carpenter's levels and plumbs shall consist essentially of a stock, a single level vial, single plumb vial, and a corresponding number of vial holders. They shall conform to table IV for the length specified (see 6.2) and shall be similar to figure 6. The level vial shall be located as shown in figure 6 and shall be protected with a metal plate.

TABLE IV. Type I, class 2, style A, adjustable vial, and style B, fixed vial, wood, top reading, carpenter's levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
12	2	2-3/8	7/8	1-1/8	7/8
18	2	2-3/8	7/8	1-1/8	1-1/8
24	2-3/8	2-3/4	1-1/16	1-1/4	2-1/4

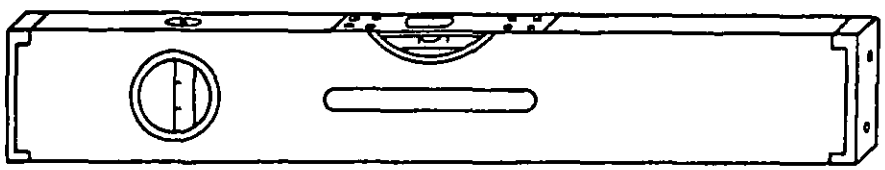


FIGURE 6. Type I, class 2, style A, adjustable vial, wood, top reading, carpenter's level and plumb.

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3.9.3.2.1 Stock. Unless otherwise specified (see 6.2), levels and plumbs of this style shall have a stock made of carefully selected, well seasoned mahogany, cherry, or sugar pine, having a clear straight grain. The stocks shall be grooved so that they may be securely gripped under service conditions. Unless otherwise specified (see 6.2), the stocks shall be protected with solid aluminum, aluminum alloy, brass, or steel end plates.

3.9.3.2.2 Vials. Level vials and plumb vials shall be adjustable.

3.9.3.2.3 Vial holders. Vial holders shall be of brass, aluminum, zinc alloy, plastic or steel. Steel parts shall be coated to prevent corrosion. They shall be so mounted as to insure side reading of the plumb vial and top reading of the level vial.

3.9.3.3 Style B, fixed vial. Style B, fixed vial, wood, top reading, carpenter's levels and plumbs shall conform to the requirements for type I, class 2, style A levels, except that the level vials and plumb vials shall be fixed and end plates are not required. They shall conform to table IV for the length specified (see 6.2) and shall be similar to figure 7.

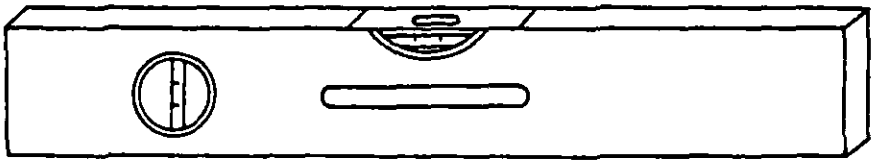


FIGURE 7. Type I, class 2, style B, fixed vial, wood, top reading, carpenter's level and plumb.

3.9.4 Class 3, wood, side reading.

3.9.4.1 Accuracy. Levels of class 3 accuracy shall conform to the requirements for class 2 levels.

3.9.4.2 Style A, adjustable vial. Style A, adjustable vial, wood, side reading, carpenter's levels and plumbs shall consist essentially of a stock with three round or oblong openings suitably located and of proper size each to accommodate one pair of adjustable twin vials. The pair nearest the center shall be levels, the other two pair shall be plumbs. All vials shall be side reading. They shall conform to table V for the length specified (see 6.2) and shall be similar to figure 8.

TABLE V. Type I, class 3, style A, adjustable vial, wood, side reading, carpenter's levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
24	2-5/16	2-7/16	1-1/16	1-3/16	1-7/8
28	2-5/16	2-7/16	1-1/16	1-3/16	2-1/8

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FIGURE 8. Type I, class 3, style A, adjustable vial, wood, side reading, carpenter's level and plumb.

3.9.4.2.1 Stock. Unless otherwise specified (see 6.2), levels and plumbs of this style shall have a stock accurately made of carefully selected mahogany, cherry, or sugar pine, well seasoned, and having a clear, straight grain. The stocks shall be grooved so that they may be securely gripped under service conditions. Unless otherwise specified (see 6.2), the stocks shall be protected with solid aluminum, aluminum alloy, brass, or steel end plates.

3.9.4.2.2 Vials. Level vials and plumb vials shall be adjustable.

3.9.4.2.3 Vial holders. Vial holders shall be of brass, aluminum, zinc alloy, plastic, or steel. Steel parts shall be coated to prevent corrosion. They shall be so mounted as to insure side reading of vials.

3.9.4.3 Style B, fixed vials. Style B, fixed vials, wood, side reading, carpenter's levels shall conform to the requirements of style A, except that the vials shall be fixed, plumbs need not be twin, or metal end plates are not required. When specified (see 6.2), plumb vials may be set 45 degrees with the base. Levels of this style shall conform to table VI for the length specified (see 6.2) and be similar to figure 9.

TABLE VI. Type I, class 3, style B, fixed vial, wood, side reading, carpenter's levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
12	2-3/16	2-5/16	13/16	15/16	1/2
18	2-5/16	2-7/16	1-1/16	1-3/16	1
24	2-5/16	2-7/16	1-1/16	1-3/16	1-1/4
28	2-5/16	2-7/16	1-1/16	1-3/16	1-5/8



FIGURE 9. Type I, class 3, style B, fixed vial, wood, side reading, carpenter's level and plumb.

3.9.5 Wood levels, brass bound. When specified (see 6.2), type I, classes 2 and 3 levels shall have the edges of the utile surfaces bound with brass binding similar to figure 10 and shall be in accordance with all other requirements of wood levels as specified herein. The brass binding shall have a minimum thickness of 0.020 of an inch and shall be securely fastened to the wood stock in such a manner so as not to loosen under normal usage.

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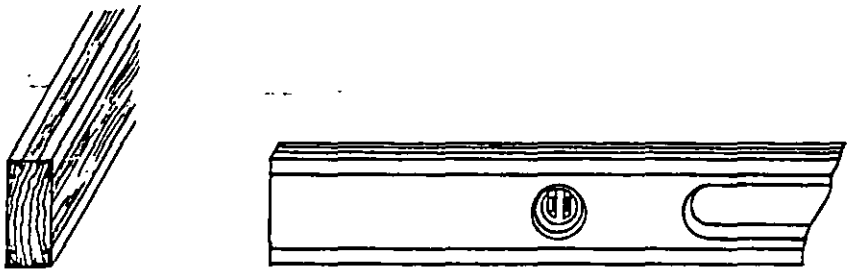


FIGURE 10. Method of brass binding for type I, classes 2 and 3 levels.

3.10 Type II, bench, precision (level and plumb).

3.10.1 Class 1, single face, fixed vial.

3.10.1.1 Style A, machinery. Style A, machinery, single face, bench levels and plumbs shall consist essentially of a stock, a level vial and two plumb vials. They shall conform to table VII for the length specified (see 6.2) and shall be similar to figure 11.

TABLE VII. Type II, class 1, style A, single face, machinery, bench levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
4	1-1/4	1-5/16	21/32	11/16	1/2
6	1-5/8	2	13/16	7/8	3/4
9	1-5/8	2	13/16	1	1-1/2
12	1-3/4	2-5/16	7/8	1	2-1/2
18	2	2-5/16	7/8	1	3-1/4
24	2	2-5/16	7/8	1	4-1/2

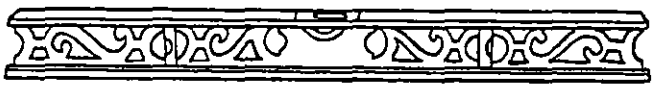


FIGURE 11. Type II, class 1, style A, single face, machinery, bench level and plumb.

3.10.1.1.1 Stock. Levels and plumbs of this style shall have a stock of close-grain, cast iron of a truss-type design to provide strength and rigidity. The base face shall be provided with a longitudinal groove through its entire length to provide a true seat for cylindrical work. Sufficient flat margin on each side of groove shall be provided for flat work.

3.10.1.1.2 Vials. The vials for class 1, single face levels and plumbs shall be precision or commercial bent-glass tubes and shall be permanently marked with two black bubble defining lines. The vials shall be bent to a uniform radius of 11.50 inches, minimum, equivalent to a vial sensitivity of 30 minutes maximum except that the main level vial of the 4 inch long size and the plumb vials of all sizes shall be bent to a uniform radius of 7.645 inches, minimum, equivalent to a sensitivity of 45 minutes, maximum. The vials shall be permanently and rigidly fixed in the stocks and shall be capable of passing the tests specified in section 4 for type II levels.

3.10.1.1.3 Accuracy. Class 1, single face levels and plumbs shall conform to table VIII for the accuracy for vial setting "a", concavity "b", as illustrated in figures 1 and 2 respectively, and the convexity requirements of 3.5.2.

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TABLE VIII. Tolerances for levels and plumbs, and levels with 30 minute vials.

Length of level "L"	Maximum tolerance	
	a	b
4 or less	0.002	0.002
6	.0025	.002
8	.0035	.002
9	.004	.003
10	.004	.003
12	.005	.004
15	.0065	.004
18	.008	.004
24	.010	.004

3.10.2 Class 2, single face.

3.10.2.1 Style A, machinists. Class 2, style A levels and plumbs shall consist essentially of a stock, adjustable level vial, vial protector, fixed plumb vial, and fixed cross test vial, except a cross test vial is not required for the 4 inch long size, and plumb vials are required only for the 12 and 18 inch long sizes. The 12 inch long size shall have one fixed plumb vial and the 18 inch long size shall have two fixed plumb vials. Style A levels and plumbs shall conform to table IX for the length specified (see 6.2) and shall be similar to figure 12.

TABLE IX. Type II, class 2, style A, single face, adjustable vial with cross test, bench levels and plumbs.

Length nominal	Width of base		Weight
	Minimum	Maximum	Maximum
Inches	Inches	Inches	Pounds
4	7/8	1	1
6	1	1-3/16	2
8	1-1/8	1-5/16	3
12	1-3/8	1-3/4	5
18	1-1/2	1-3/4	6



FIGURE 12. Type II, class 2, style A, single face, adjustable vial with cross test, bench level and plumb.

3.10.2.1.1 Stocks. Stocks of this style shall be made of cast iron. The base shall be provided with a vee or concave groove through its entire length. Sufficient flat margin on each side of groove shall be provided for a good seat for flat work. The groove shall be straight and shall provide a true seat for cylindrical work.

3.10.2.1.2 Vials. The main level vial for style A shall be a lime or heat-resistant boro-silicate glass tube and shall be ground to an inside radius of 229.3 inches, minimum, equivalent to a vial sensitivity of 90 seconds, maximum, except the level vial of the 4 inch long size shall be ground to an inside radius of 34.40 inches minimum, equivalent to a 10 minute vial sensitivity maximum. There shall be no less than two auxiliary lines, permanently marked, spaced 1/10 inch apart and 1/10 inch from the outside of both bubble defining lines, and the value of these lines shall be permanently marked on the levels and plumbs of the vial protector or holder. The main level vial shall be adjustable within the accuracy tolerance dimension "a" of table X as illustrated by figure 1. The plumb and cross

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test vial shall be fixed in the stock and shall conform to 3.9.1. The vials shall be capable of passing the tests specified in section 4 which are applicable to type II, class 2 levels.

TABLE X. Tolerances for styles A and B levels and plumbs.

Length of level "L"	Maximum tolerance	
	a	b
4	0.0005	0.002
6	.0005	.002
8	.001	.002
12	.001	.002
18	.0015	.002

3.10.2.1.3 Accuracy. Class 2, style A, single face levels and plumbs shall conform to table X for the accuracy of the main vial setting "a", and for concavity "b" as illustrated in figures 1 and 2 respectively, and the convexity requirements of 3.5.2. The accuracy for the vial setting "a" of the plumb and cross test vial shall conform to table I. The levels and plumbs shall be capable of passing the tests specified in section 4 for type II, class 2 levels.

3.10.2.1.4 Vial protector. The holder for the main level shall be equipped with means for protecting the vial from damage. This device shall be readily accessible and easily operated.

3.10.2.2 Style B, millwright's. Style B levels and plumbs shall conform to type II, class 2, style A, for essential component parts.

3.10.2.2.1 Vials. The main level vials shall be ground or precision bent-glass tubes and shall have an inside radius of 34.40 inches, minimum, equivalent to a vial sensitivity of 10 minutes, maximum. There shall be no less than one auxiliary line spaced 1/10 inch on the outside of each bubble defining line. The main level vial shall be adjustable within the accuracy tolerance "a" of table X as illustrated by figure 1. The plumb and cross test vials shall conform to the requirements for type I level vials. The vials shall be capable of passing the tests specified in section 4 as applicable to type II, class 2, style A level vials.

3.10.2.2.2 Accuracy. Style B levels and plumbs shall conform to the accuracy requirements for style A or type II, class 2.

3.11 Type III, mason's.

3.11.1 Vials. Vials for type III mason's levels and plumbs shall be 45 minute sensitivity vials and shall conform to the requirements for type I level vials.

3.11.2 Class 1, metal, side reading.

3.11.2.1 Accuracy. Accuracy for class 1, metal, side reading, mason's levels and plumbs shall conform to the requirements for type I, class 1 level accuracy.

3.11.2.2 Style A, adjustable vial. Style A levels and plumbs shall consist essentially of a stock with three or more round or oblong openings, each to accommodate side reading twin adjustable vials. The openings nearest the center shall contain level vials, the others to contain plumb vials. They shall conform to table XI and be similar to figure 13.

TABLE XI. Type III, class 1, style A, adjustable vial, metal, mason's levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
48	2-1/4	2-5/8	1	1-5/16	5

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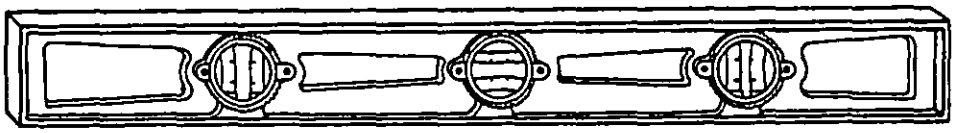


FIGURE 13. Type III, class 1, style A, adjustable vial, metal, mason's level and plumb.

3.11.2.2.1 Stocks. Stocks shall be of "I" beam construction, of aluminum alloy or magnesium of adequate strength and rigidity for the purpose intended. The levels and plumbs shall be tested for strength in accordance with 4.5.3.6.

3.11.2.2.2 Vial holders. Vial holders shall be of aluminum, aluminum alloy, zinc alloy, plastic, or steel so mounted as to assure side reading. Steel parts shall be coated to prevent corrosion.

3.11.2.3 Style B, fixed vial. Fixed vial side reading, metal, mason's levels and plumbs shall conform to the requirements for type III, class 1, style A, except the vials shall be fixed. Style B levels shall conform to table XII and shall be similar to figure 14. The levels and plumbs shall be tested for strength in accordance with 4.5.3.6.

TABLE XII. Type III, class 1, style B, fixed vial, metal, side reading, mason's levels and plumbs.

Length nominal	Depth		Width Minimum	Weight Maximum
	Minimum	Maximum		
Inches	Inches	Inches	Inch	Pounds
48	2-1/4	2-5/8	1	5



FIGURE 14. Type III, class 1, style B, fixed vial, metal, side reading, mason's level and plumb.

3.11.3 Class 2, wood, side reading.

3.11.3.1 Accuracy. Accuracy for class 2, wood, side reading, mason's levels and plumbs shall conform to the requirements for type I, class 3, style A levels.

3.11.3.2 Style A, adjustable vial. Style A, adjustable vial levels and plumbs shall consist essentially of a stock with three or more round or oblong openings suitably located and of proper size, each to accommodate one pair of adjustable vials. The pair nearest the center shall be levels, the others shall be plumbs. All vials shall be side reading. They shall conform to table XIII and shall be similar to figure 15.

TABLE XIII. Type III, class 2, style A, adjustable vial, wood, side reading, mason's levels and plumbs.

Length nominal	Depth		Width Minimum	Weight Maximum
	Minimum	Maximum		
Inches	Inches	Inches	Inch	Pounds
48	2-5/16	2-7/16	1-1/8	3-1/8

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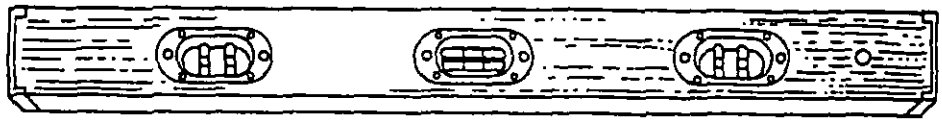


FIGURE 15. Type III, class 2, style A, adjustable vial, wood, side reading, mason's level and plumb.

3.11.3.2.1 Stocks. Unless otherwise specified (see 6.2), levels and plumbs of this style shall have a stock made of carefully selected cherry, mahogany, black walnut, or sugar pine, well seasoned and having a slope of grain not greater than 1 in 15. A hole suitable for hanging shall be provided in accordance with manufacturing standards. Stocks shall have nonferrous metal bound ends and, unless otherwise specified, the edges may be plain or metal bound. The sides of the stock shall be grooved so that the tool may be securely gripped under service conditions.

3.11.3.2.2 Vial holders. Vial holders shall be of brass, aluminum or zinc alloy, plastic, or steel. Steel parts shall be coated to prevent corrosion. They shall be so mounted as to insure side reading of the vials.

3.11.3.3 Style B, fixed vial. Style B, fixed vial levels and plumbs shall conform to the requirements for type III, class 2, style A, except the openings may be oval or oblong and the vials shall be fixed. Style B levels and plumbs shall conform to table XIV and shall be similar to figure 16.

TABLE XIV. Type III, class 2, style B, fixed vial, wood, side reading, mason's levels and plumbs.

Length nominal	Depth		Width		Weight Maximum
	Minimum	Maximum	Minimum	Maximum	
Inches	Inches	Inches	Inch	Inches	Pounds
48	2-1/4	2-1/2	1-1/16	1-1/4	3-1/16



FIGURE 16. Type III, class 2, style B, fixed vial, wood, side reading, mason's level and plumb.

3.12 Type IV, torpedo (pocket). Torpedo pocket levels and plumbs shall consist essentially of a stock, one level vial, one plumb vial, and one 45 degree vial. They shall be similar to figure 17 and shall be of a nominal length of 9 inches.

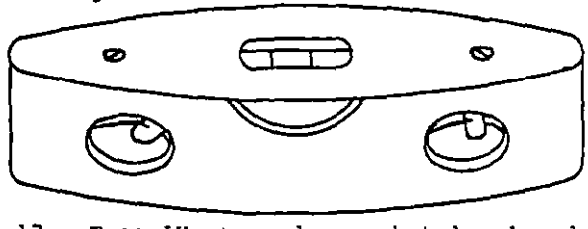


FIGURE 17. Type IV, torpedo, pocket level and plumb.

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3.12.1 Vials. Vials for type IV, torpedo levels and plumbs shall conform to the requirements for type I, except the plumb vial and 45 degree vial shall be bent to a radius of 5.8 inches, minimum, equivalent to a vial sensitivity of 1 degree. The vials shall be fixed.

3.12.2 Accuracy. The maximum tolerance for the "a" dimension of figure 1 shall be 0.009 inches and for the "b" dimension of figure 2 shall be 0.005 inches and shall conform to the convexity requirements of 3.5.2. The level and plumb shall be capable of passing the applicable tests specified in section 4.

3.12.3 Stocks. Stocks for torpedo levels shall be of wood, aluminum, or magnesium or as specified (see 6.2). The base shall be flat and smoothly finished. The base shall have a "V" or concave groove throughout its entire length.

3.13 Type VI, bench (level).

3.13.1 Class 1, single face.

3.13.1.1 Style A, fixed vial. Style A, fixed vial level shall consist essentially of a stock and one level vial. They shall conform to table XVI for the length specified (see 6.2) and shall be similar to figure 20.

TABLE XVI. Type VI, class 1, style A, single face, fixed vial, bench level.

Length nominal	Width of base minimum	Weight maximum
Inches	Inch	Ounces
3-1/2	1/2	6
4	1/2	8
5	1/2	10
6	1/2	12

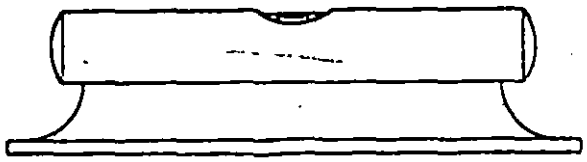


FIGURE 20. Type VI, class 1, style A, single face, fixed vial, bench level.

3.13.1.1.1 Vials. Vials shall be fixed and shall conform to the requirements for type I level vials.

3.13.1.1.2 Stocks. Levels of this style shall have a stock of iron, cast iron, or steel, the base of which shall be smoothly finished.

3.13.1.1.3 Accuracy. Accuracy for class 1 levels shall conform to the requirements for type I, class 1, levels vials.

3.13.2 Class 2, straightedge. Class 2, straightedge bench levels shall consist of a stock, one level vial, and means for attaching to a straightedge. Levels of this class shall be similar to figure 21 and shall be of one size, approximately 3 inches long.

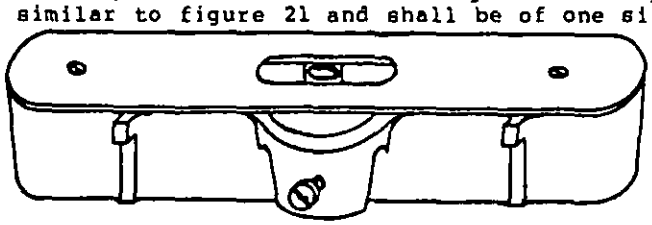


FIGURE 21. Type VI, class 2, straightedge, bench level.

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3.13.2.1 Stocks. Stocks shall conform to type VI, class 1 requirements.

3.13.2.2 Vials. Vials shall be fixed and conform to the requirements for type VI, class 1.

3.13.2.3 Accuracy. Accuracy for type VI, class 2 levels shall conform to the requirements for type I, class 1 level vials.

3.13.3 Class 3, cross test. Class 3, cross test bench levels shall consist essentially of a stock and two level vials. They shall conform to table XVII for the length specified (see 6.2) and shall be similar to figure 22.

TABLE XVII, Type VI, class 3, cross test, bench level.

Length of one arm, nominal	Length of other arm, nominal	Weight maximum
Inches	Inches	Ounces
1	1-3/8	2
1-1/4	1-1/4	2
2	3	3
2-3/4	2-3/4	4

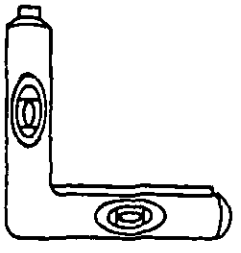


FIGURE 22. Type VI, class 3, cross test, bench level.

3.13.3.1 Stocks. Levels of this type shall have a stock of iron, cast iron, or steel designed with two arms at 90 degrees to each other on a horizontal plane. Each arm shall have a base pad at right angles to each other and shall be finished smooth and flat so as to provide an accurate bearing surface. In addition to the base pad, each arm shall also support a top reading level vial.

3.13.3.2 Vials. Vials shall be fixed and shall conform to the requirements for type VI, class 1 levels.

3.13.3.3 Accuracy. Accuracy for type VI, class 3 levels shall conform to that for type VI, class 1 levels.

3.13.4 Class 5, striding. Class 5 striding levels shall consist essentially of a brass stock, one level vial, and a vial protector. They shall be similar to figure 23 and have nominal dimensions for length 5-3/16 inches, width 1 inch, height 1-7/8 inches. The distance between the legs shall be 4-1/4 inches. The dimensional tolerances shall be ± 1/16 inch.

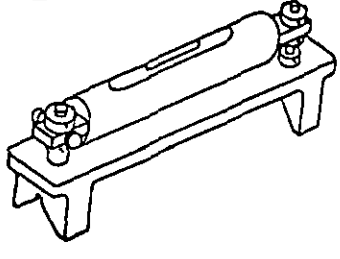


FIGURE 23. Type VI, class 5, striding bench level.

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3.13.4.1 Vial. The level vial shall be adjustable within the accuracy tolerance, dimension "a" of table X, 12 inch size, and shall conform to the requirements for type II, class 2, style A level vials.

3.13.4.2 Vial protector. The vial protector shall conform to the requirements for type II, class 2, style A level vial protectors.

3.13.4.3 Accuracy. Class 5 levels shall conform to the requirements for the 12 inch size type II, class 2, style A levels for accuracy.

3.14 Type VII, midget (pocket). Midget pocket levels shall consist essentially of a stock and one level vial, shall conform to table XIX for the length specified (see 6.2) and shall be similar to figure 24.

TABLE XIX. Type VII, midget (pocket) levels.

Length + 1/32	Width of base minimum	Weight maximum
Inches	Inch	Ounces
2	13/64	1
3*	13/64	2
4	13/64	2-1/4
5	13/64	2-1/4
6	13/64	3

*Minimum 2-1/2, maximum 3-1/2 inches

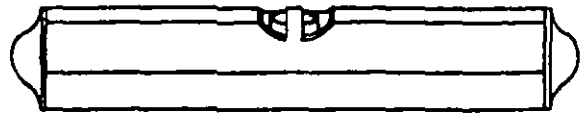


FIGURE 24. Type VII, midget (pocket) level.

3.14.1 Stocks. Levels of this style shall have a stock of plated brass or aluminum alloy of a square, hexagonal or octagonal cross section, with appropriate end caps. The stock may have an optional grooved base or pocket clip, as specified (see 6.2).

3.14.2 Vials. The level vial shall be fixed and conform to the requirements for type I levels, except that a centering bar, an integral part of the stock, may be employed in place of bubble defining lines marked on the vial.

3.14.3 Accuracy. Type VII levels shall conform to the requirements for type I levels and table I for the "a" and "b" dimensions, and the convexity requirements of 3.5.2.

3.15 Type VIII, line. Type VIII line levels shall consist of a fixed level vial and a stock with hooks for attaching to a line. Levels of this type shall be similar to figure 25 and shall have a nominal length of 3 inches. The weight of the level shall not exceed 1/2 ounces.



FIGURE 25. Type VIII line level.

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3.15.1 Vials. The level vials shall be fixed and shall conform to the requirements for type I level vials.

3.15.2 Stocks. Levels of this type shall have a stock of brass, aluminum, or aluminum alloy. The bottom surface shall be smooth and flat, capable of being used as a utile surface. Stocks shall be provided with end plate hooks for attaching to the line.

3.15.3 Hooks. Hooks shall be integral with the end plates and shall be sufficiently rugged to withstand rough handling and storage without deformation. Hooks shall be so designed as to prevent accidental detachment and so positioned as to suspend the level parallel to the line.

3.15.4 Accuracy. Type VIII levels shall conform to the requirements for type I level accuracy.

3.16 Type X, circular levels (mounted). Circular levels shall have a glass vial mounted to a metal cylinder and shall be similar to figure 27. They shall conform to table XXII for the size vial specified (see 6.2).

TABLE XXII. Circular levels

Vial size diameter	Cylinder diameter		Height ± 1/32	Vial sensitivity
	Maximum	Minimum		
Inches	Inches	Inches	Inch	(maximum)
1/2	1	7/8	3/8	45 minutes
5/8	1-5/32	1	7/16	45 minutes
3/4	1-3/8	1-1/4	7/16	45 minutes
7/8	1-9/16	1-7/16	33/64	20 minutes
1	1-27/32	1-3/4	5/8	10 minutes
1-1/4	2-1/4	2	11/16	10 minutes
1-1/2	2-3/4	2-1/4	11/16	10 minutes
2	3	2-3/4	11/16	10 minutes
2-1/2	2-7/8	2-5/8	11/16	10 minutes

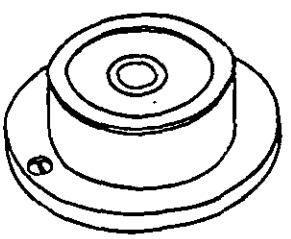


FIGURE 27. Type X, circular level (mounted).

3.16.1 Stocks. Unless otherwise specified (see 6.2), circular levels shall be made of ferrous or nonferrous metal. The base shall be flat and smoothly finished. Each level furnished shall be so adjusted that when the bubble is centered, the base of the cylinder is horizontal. A circle shall be inscribed on the level for centering the bubble.

3.16.2 Vials. The vial shall be fixed in the level. When specified (see 6.2), the exterior side of the vial shall be coated with a radioactive luminous material. The vial shall have a bubble defining reference circle with a diameter normally accepted by industry, equal to 30 to 40 percent of the diameter of the vial. Vials shall be commercial glass and shall possess sufficient strength to withstand the stresses produced by sudden temperature changes and the tests specified in section 4. The vial sensitivity shall conform to table XXII for the diameter specified (see 6.2). The vials shall be capable of passing the test specified in section 4.

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3.17 Marking. Each level shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined, part number, and the country of origin.

3.18 Workmanship. Levels shall conform to the quality and grade of product described in the requirements of this specification. Levels shall be free from rust and defects which may impair their serviceability, durability or appearance.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance inspection.

4.2.1 Lot. All levels and plumbs of the same type, class, style, and size offered for delivery at one time shall be considered a lot for purposes of quality conformance inspection.

4.2.2 Sampling. A random sample of levels and plumbs and levels shall be selected from each lot in conformance with MIL-STD-105 at inspection level II.

4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each sample selected in accordance with 4.2.2. Quality conformance inspection shall consist of the following:

(a) Examination (4.4).
(b) Tests (4.5).
(c) Inspection and preparation for delivery (4.6).

4.4 Inspection.

4.4.1 Examination. Each of the sample levels and plumbs and levels selected in accordance with 4.2.2 shall be examined by the inspector to verify compliance with this specification. Examination shall be conducted as specified in table XXIII. Any level and plumb, and level in the sample containing one or more defects shall be rejected, and if the number of defective levels and plumbs and levels in any sample exceeds the accepted number for that sample, the lot represented by the sample shall be rejected. Particular attention shall be given to examination for the following characteristics:

(a) Blowholes, cracks, or splits in metal stock.
(b) Checks, cracks, or splits in wood stock.
(c) Impractical depth of hand groove.
(d) Protruding end or cover plate.
(e) Loose vials, or vial holder lenses.
(f) Inadequate visibility of bubbles.
(g) Illegible or improper marking.
(h) Finish of part.
(i) Operating difficulties of adjustments.
(j) Poor workmanship.
(k) Sharp edges where not required.
(l) Improper number of vials.
(m) Vial holders not easily replaceable.
(n) Vials unprotected from damage.

The acceptable quality level (AQL) shall be 1.5 percent defective for major defects and 2.5 percent for minor defects.

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TABLE XXIII. Classification of defects.

Categories	Defects
Critical	None defined.
Major	
101	Type, class, style, and size not as specified.
102	Stock material not as specified; not of specified construction; not of truss-type construction, evidence of deep pits, cracks, porosity, blowholes, nicks, scratches, dents, seams, burrs, or warping (except type I, classes 2 and 3, styles A and B and type III, class 2, styles A and B).
103	Stock not of the specified species of wood; evidence of cracks, checks, splits, soft spots, or heavy knots, grain slope greater than allowable (type I, classes 2 and 3, styles A and B; type III, class 2, styles A and B only).
104	Stock surfaces not smoothly finished; not coated with a suitable material to prevent moisture absorption (type I, classes 2 and 3, styles A and B and type III, class 2, style A and B only).
105	Stock base or arms, as applicable, not smoothly finished (type VI, class 1, style A, classes 2 and 3 only).
106	Stock not tubular construction (type VII only).
107	Stock configuration not as specified (type VII only).
108	Stock ends not protected with metal plates, end plates not securely fastened (type I, class 3, style A and type III, class 2, style A only).
109	End plate material not as specified; not smoothly finished and polished; when applicable, not coated (steel), (type I, class 2, style A, class 3, style A and type III, class 2, style A)
110	End caps on end plates missing, damaged, or not securely fastened to stock.
111	Stock base not grooved, groove not straight (type VII only, when required).
112	Groove in stock missing, insufficient depth and length (type I, classes 2 and 3, styles A and B, type III, class 2, styles A and B and type IV).
113	Groove, vee or concave not provided through entire length of base of stock, not straight (type II, class 2, style A only).
114	Arms not 90 degrees to each other on a horizontal plane (type VI, class 3 only).
115	Unmachined surfaces not coated as required, evidence of bare spots, peeling, or flaking (type II, class 1, style A; type II, class 2, style A; type VI, class 1, style A, classes 2 and 3.
116	Number and type (level or plumb) of vials not as required, improperly located, not clearly visible (type I, class 1 only).
117	Number and type (level or plumb, single or double) of vials not as required, improperly located, not clearly visible (type I, classes 2 and 3, styles A and B and type III, class 2, styles A and B only).
118	Number and type (level, plumb, or cross test) of vials not as required, improperly located, not clearly visible (type II, class 2, style A only).
119	Number and type (level, plumb, 45 degrees) of vials not as required, improperly located, not clearly visible, not firmly set in stock (type IV, only).
120	Insufficient flat surface on each side of groove (type II only).

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Categories	Defects
121	Longitudinal groove not provided through entire length of base, face of stock not clearly visible (type II, class 1 only).
122	Straight edge attaching device missing, damaged, or nonconforming (type VI, class 2 only).
123	Pocket clip missing, damaged, improperly assembled to stock (type VII only).
124	Plate missing, evidence of bare spots or peeling (type VII brass only).
125	Hooks missing, not integral with end plates, damaged, bent, or improperly positioned (type VIII only).
126	Vial(s) improperly located, not clearly visible, not firmly set in stock (type VI, class 1, style A, classes 2 and 3 only).
127	Vial material not as specified; vials cracked, fogged, not clear; bubble centering lines missing or not permanently marked; incorrectly located or less than the number required; not firmly set stock.
128	Vial protectors missing or damaged, not securely mounted, interfere with vial visibility or working surface; not readily accessible, does not operate easily; material not as required.
129	Vial holders not of the specified material; plating missing or incomplete, not interchangeable as required, not secured in stock, vials not firmly held, not adjustable.
130	Vial holder (main level vial) not of the required material; coating missing or incomplete, evidence of bare spots, peeling or flaking, not interchangeable as required (type II, class 2 only).
131	Vial holder (main level vial) not adjustable, cannot be locked in fixed position, not securely attached to stock, vial not firmly held or protected (type II, class 2 only).
132	Stock surfaces, and vial holders, when applicable, not smooth and polished (type I, class 1 only).
133	Plumb vial(s), when applicable, and cross test vial not firmly set in stock (type II, class 2, style A only).
134	Width of base not as required, not within tolerance (type II, class 2, style A, and type VI, class 2 only).
135	Length of arms not as specified, not within tolerance (type VI, class 3 only).
136	Length not as specified, not within tolerance.
137	Depth not within specified limits (type I, classes 1, 2, and 3; type II, class 1; type III, classes 1 and 2 only).
138	Width not within specified limits or not within tolerance (type I, classes 1, 2, and 3; type II, class 1; type III, class 2, styles A and B).
Minor	
201	Handhold missing in stock (type I, classes 2 and 3, type III, class 2 and type VI, classes 1, 2, and 3 only).
202	Marking, manufacturer's name or trademark, country of origin, type, class, specification, manufacturer's part number, incorrect or illegible.

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4.4.2 Quality conformance tests. Each sample level and plumb, and level selected in accordance with 4.2.2 shall be tested to verify compliance with this specification. Any level and plumb, and level in the sample failing to pass one or more of the tests shall be rejected and if the number of defective levels and plumbs, and levels in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. The AQL shall be 1.5 percent defective for tests defects. Each sample level and plumb, and level shall be tested for the following:

(a) Nonconformity to vial setting tolerance.
(b) Excessive concavity or convexity.
(c) Improperly marked cross lines.
(d) Groove in base not parallel with base.
(e) Special tests specified in 4.5, as applicable.

4.5 Test procedures.

4.5.1 Test for type I, class 1; type II, classes 1 and 2; type III, class 1; type IV; type VI, classes 1, 2, and 3; type VII, class 1; and type VIII.

4.5.1.1 Parallelism test. The depth of stock shall be measured at two points at each end to determine the parallelism along and across the width, which shall not vary more than that specified for dimension "c" (see 3.9.2.1).

4.5.1.2 Concavity and convexity test. Each utile surface of the level and plumb or level shall be supported, at the extreme ends, by two precision gage blocks or parallels placed on a precision surface plate. The entire utile surface supported by the gage block or parallels shall be indicated with a precision indicator. The indicator shall not show any convexity and shall not show a variation, for the "b" dimension on figure 2, greater than the "b" tolerance specified, in the applicable tables I, VIII, X, and paragraph 3.12.2.

4.5.1.3 Level vial setting test. A precision surface plate or precision parallel, that has its entire utile surface leveled in a true horizontal plane by use of a precision level of known accuracy, shall be used to test the level vial setting of levels and plumbs and levels. The thickness of the gage required under one end of the level and plumb or level to center the bubble between the bubble defining lines of the vial shall not exceed the "a" tolerance specified in tables I, VIII, X, and paragraph 3.12.2 as illustrated for the "a" dimension of figure 1. The gage shall not extend more than 3/8 inch under the end of the level stock.

4.5.1.4 Plumb vial setting test. The plumb vial setting shall be tested in accordance with 4.5.1.3 except that a precision angle plate shall be attached to the surface plate to provide a true vertical plane for testing the plumb vial setting. Care shall be taken to insure that the sides of the level and plumb are in an approximate vertical plane.

4.5.1.5 Vial-sensitivity test for level vials. The level and plumb or level shall be placed on a parallel bar of known accuracy for parallelism and straightness of the two utile surfaces. The parallel bar shall be either 12, 24, or 48 inches in length and shall be supported, at the extreme ends, by precision gage blocks located on a precision surface plate as shown on figure 28. The bubble shall be centered between the bubble defining lines as described in 4.5.1.3. The height of the gage blocks at one end of the parallel shall be increased to cause a bubble movement of 1/10 inch. This increase in height shall not exceed that specified in table XXIV for the vial tested. For the purpose of this test, the gage blocks shall be positioned not more than 3/8 inch under the end of the parallel bar.

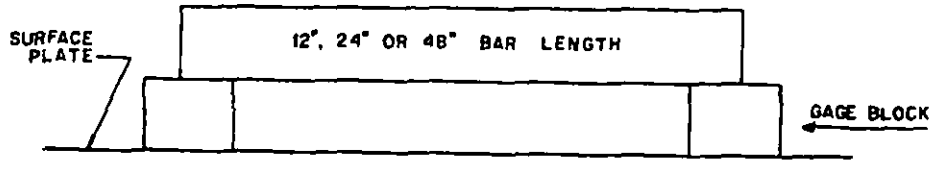


FIGURE 28. Parallel bar and gage blocks for determining vial sensitivity.

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TABLE XXIV. Vial sensitivity test.

Degree of vial sensitivity		Maximum increase in the height of the gage block at one end of the parallel required for a 1/10 inch bubble movement.		
		Length of parallel bar		
		1 foot	2 feet	4 feet
Minutes	Seconds	Inch	Inch	Inch
--	10	0.0006	0.0012	0.0024
--	90	.0054	.0108	.0216
4	--	.014	.028	.056
10	--	.035	.070	.140
20	--	.070	.140	.280
25	--	.088	.175	.350
30	--	.105	.210	.420
45	--	.157	.314	.628
86	--	.300	.600	.200

4.5.1.6 Vial-sensitivity test for plumb vials. The plumb vial sensitivity shall be tested in accordance with 4.5.1.5 except that a precision angle plate shall be attached to the surface plate to provide a true vertical plane for testing plumb vials. Care shall be taken to insure that the sides of the level and plumb and also in an approximate vertical plane.

4.5.2 Test procedures for type I, classes 2 and 3; type III, class 2; and type X.

4.5.2.1 Convexity and concavity test. The test, except for type X, shall be conducted in accordance with 4.5.1.2 except that the total indicator variation shall not exceed 0.015 inch for each utile surface or show any convexity while the level and plumb remains in one position. Type X levels do not require this test.

4.5.2.2 Level-vials setting test. The test, except for type X, shall be conducted in accordance with 4.5.1.3. Type X circular levels shall be located on a precision surface plate horizontally level within 0.001 inch per foot in both the longitudinal and transverse direction. The amount of adjustment required to center the bubble in the circle inscribed on the vial shall not exceed 0.007 inch per foot.

4.5.2.3 Plumb vial setting test. The test, except for type X, for plumb vial setting, shall be conducted in accordance with 4.5.1.4.

4.5.2.4 Vial-sensitivity test for level vials. The test for type I, classes 2 and 3; type III, class 2, shall be conducted in accordance with 4.5.1.5. The vial sensitivity of the type X circular level shall be tested for compliance with table XXII.

4.5.2.5 Vial-sensitivity test for plumb vials. The plumb vial sensitivity shall be tested in accordance with 4.5.1.6.

4.5.3 Special tests.

4.5.3.1 Forty-five degree test. All level and plumb or level vials set at 45 degrees with the base shall, in addition to tests of 4.5, be tested by positioning the base upon a sine plate or bevel protractor set to 45 degrees. The bubble of the 45 degree vial shall meet the requirements for dimension "a" of figure 1, when tested in accordance with 4.5.1.3.

4.5.3.2 Line-level test. Type VIII line levels shall be tested in addition to the tests of 4.5.1 by suspension from a taut line supported on two equal spacer blocks above the test plate of 4.5.1.3. Dimension "a" in figure 1 shall be determined by the distance between the line support blocks and not the level length.

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4.5.3.3 Grooved base test. Levels having a longitudinal groove through the entire length shall be tested by placing the groove of the base on an accurate cylindrical bar resting on the test plate. Procedure shall be similar to 4.5.1.3.

4.5.3.4 Temperature test. One type X level from each sample of type X levels shall be placed in a temperature test chamber and held at a temperature of not less than 125 degrees F for a period of 3 hours, and then placed in a temperature test chamber and held at a temperature of -65 degrees F for a period of 3 hours. Circular levels shall withstand the sudden temperature changes of this test without sufficient loss of accuracy to fail the vial accuracy test of 4.5.2.2 and without developing other disqualifying defects.

4.5.3.5 Vial temperature test. The vials of one sample level of types I through VIII shall be placed in a temperature test chamber and held at a temperature of +120 degrees F for types II and VI levels, and at a temperature of +150 degrees F for types I, III, IV, VII, and VIII levels for 8 hours, and removed from the chamber and allowed to return to an ambient temperature of +60 degrees F to +90 degrees F. The vials shall then be placed in a temperature test chamber and held at a temperature of -30 degrees F for a period of 8 hours. The vials shall be tested mounted in the types I, III, IV, VII, and VIII level stocks. A type I, 24 inch length level may be substituted for the 28- and 30 inch type I levels provided it is of the same quality, style and class. Vials of the type III levels shall be tested in sections meeting the vial setting and vial sensitivity tests. The sections shall be 24 inches or less in length. The vials shall be examined for the requirements of 3.7 and shall not fail the applicable tests of 4.5.1.3 through 4.5.1.6 or develop other disqualifying defects.

4.5.3.6 Loading test. One sample level of each length of the type I, class 1 and type III, class 1 levels and plumbs shall be placed across two rounded supports that provide line contact. The distance between the centers of the supports shall be 1/2 inch less than the length of the level. The load shall be applied gradually at a rate no faster than 0.2 inch per minute at the midpoint of the level and plumb using a cylinder to provide line contact. The level shall be loaded to the amount specified in table XXV and then released completely at a rate no slower than 0.2 inch per minute. The deflection at the midpoint after the load has been removed for a period of 15 minutes shall not exceed 0.005 inch. The deflection shall be measured by employing a dial indicator or measuring the change in concavity at the midpoint of the level and plumb.

TABLE XXV. Load test

Length of Level	Test Load
Inches	Pounds
18	735
24	550
28	470
30	440
48	275

4.6 Inspection of preparation for delivery. An inspection shall be made to determine that the preservation, packaging, packing, and marking requirements comply with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking shall be in accordance with PPP-P-40. The level of preservation, packaging, and packing shall be A, B, C, or as specified (see 6.2).

6. NOTES

6.1 Intended use. Levels and plumbs are used to establish a true horizontal or vertical plane, being equipped with both level and plumb vials. Levels are used to establish a true horizontal plane, being equipped with level vials only. The types I and III levels and plumbs are used by the craftsmen indicated by the respective titles. The type II levels and plumbs and the type VI levels are used mainly by machinists with the type II, class 2, style A and type IV, class 4 levels the most

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precise. The type IV levels and plumbs and the type VI levels are smaller and may be conveniently carried in a pocket. The type VIII level is used in laying foundations, tile pipe, brick walls and determining grades. The type X level is used to determine a horizontal plane in all directions on one setting.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, and style required (see 1.2).
(c) Length of level required (see applicable table).
(d) Whether special 45 degree set of plumb vial is required (see 3.9.4.3).
(e) Stock material requirements if different (see 3.12.3, 3.11.3.2.1, 3.9.4.2.1, and 3.9.3.2.1).
(f) State when end plates are not required (see 3.9.3.2.1 and 3.9.4.2.1).
(g) Whether wood levels shall be brass bound (see 3.9.5).
(h) Whether the stock of the midget pocket level shall have a grooved base or pocket clip (see 3.15.1).
(i) Diameter of vial required (see 3.18).
(j) Whether luminous coating is required (see 3.18.2).
(k) Applicable levels of preservation, packaging, and packing (see 5.1).
(l) Marking, if different (see 5.1).
(m) If other preservation, packaging, and packing are required (see 5.1).

6.3 Supersession data: The types of levels and plumbs, and levels previously covered by this specification have been superseded as follows:

GGG-L-211C	Superseded by:
Type V	DELETED
Type IX	DELETED

MILITARY INTERESTS:

Custodian:

Air Force - 99

Review activity:

Air Force - 84

PREPARING ACTIVITY:
GSA/FSS

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